



# A note on the online First-Fit algorithm for coloring $k$ -inductive graphs

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## ARTICLE INFO

### Article history:

Received 19 February 2008  
Received in revised form 29 July 2008  
Available online 23 August 2008  
Communicated by K. Iwama

### Keywords:

On-line algorithms  
Analysis of algorithms  
Graph algorithms

## ABSTRACT

In a FOCS 1990 paper, S. Irani proved that the First-Fit online algorithm for coloring a graph uses at most  $O(k \log n)$  colors for  $k$ -inductive graphs. In this note we provide a very short proof of this fact.

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## 1. Introduction

An online graph coloring algorithm  $A$ , is an algorithm that receives as input, the set of vertices of a graph in an arbitrary order in time steps. In each time step, the algorithm  $A$  is revealed a new vertex  $v$  of the graph together with all of the edges that connect  $v$  with the set of vertices that were revealed in previous time steps. The algorithm must assign  $v$  a color that is distinct from all colors assigned to neighbors of  $v$  that were already revealed. Once  $v$  is assigned a color, this color cannot be changed in the future. For a graph  $G$ , the *performance* of  $A$ , denoted  $A(G)$  is the maximum number of colors used by  $A$ , where the maximum is taken over all possible permutation inputs of the vertices of  $G$ . The *competitive ratio* of  $A$  for a given class of graphs  $\mathcal{C}$  is simply  $\max_{G \in \mathcal{C}} A(G)/\chi(G)$ , where  $\chi(G)$  is the chromatic number of  $G$ .

In this note we focus on the class of  $k$ -inductive graphs. A graph  $G = (V, E)$  is called  $k$ -inductive (or  $k$ -degenerate) if any subgraph  $H \subset G$  contains a vertex of degree at most  $k$ . Equivalently, it is  $k$ -inductive if and only if there exists a bijective numbering of its vertices with  $\{1, \dots, n\}$  (where  $n = |V|$ ) such that every vertex has at most  $k$  neighbors with lower numbers. Indeed, given such a numbering, let

$U \subset V$  and let  $H'$  be a subgraph of  $G$  with vertex set  $U$ . Then obviously the vertex of  $U$  with the largest number has at most  $k$  neighbors in  $H'$ . In the other direction, one can number the vertices recursively by choosing a vertex in  $V$  with degree at most  $k$ , number it with “ $n$ ” and continue recursively on the subgraph spanned by  $V \setminus \{v\}$ .

It is easy to see (using induction) that the chromatic number of such graphs is at most  $k + 1$ . Inductive graphs are an important sub-class of graphs. For example, it easily follows from Euler formula that planar graphs are 5-inductive and bipartite planar graphs are 3-inductive. It is easily seen that a chordal graph  $G$  is  $\chi(G)$ -inductive.

The First-Fit (FF) algorithm is an online algorithm that colors every input vertex with the smallest possible integer. We consider the First-Fit greedy algorithm for coloring graphs online. The performance of this algorithm for restricted classes of graphs has been analyzed in several papers (see, e.g., [1–3]). S. Irani considered the class of  $k$ -inductive graphs (also known as  $k$ -degenerate graphs). In [2], Irani showed that First-Fit uses at most  $O(k \log n)$  colors and thus proving that the competitive ratio of this algorithm for the class of  $k$ -inductive graphs is  $O(\log n)$ . In [2] it is also shown that this bound is tight in the sense that any online coloring algorithm has competitive ratio  $\Omega(\log n)$  on the class of  $k$ -inductive graphs.

In this note, we provide an alternative analysis for the performance of First-Fit on  $k$ -inductive graphs. Surpris-

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ingly, our analysis is very short. We believe that our analysis is also much simpler than the one provided in [2].

## 2. A simple analysis of online First Fit

Let  $G = (V, E)$  be a  $k$ -degenerate graph with  $n$  vertices and let  $FF$  denote the First-Fit online coloring algorithm. Fix an input permutation  $\sigma = \{v_1, v_2, \dots, v_n\}$ . For a vertex  $v \in V$ , let  $FF_\sigma(v)$  denote the color assigned to  $v$  by  $FF$  when running on input  $\sigma$ . Let  $C_i \subset V$  denote the set of vertices that are assigned colors greater than or equal to  $i$  by  $FF$  with input  $\sigma$ . Formally:

$$C_i = \{v \in V \mid FF_\sigma(v) \geq i\}.$$

We need the following easy facts on  $k$ -degenerate graphs:

**Lemma 2.1.** *Let  $G$  be a simple  $k$ -degenerate graph and let  $H \subset G$  be a subgraph of  $G$ . Let  $e(H)$  and  $v(H)$  denote the number of edges and of vertices of  $H$ , respectively. Then:*

- (i)  $e(H) \leq k \cdot v(H)$ .
- (ii) *The number of vertices in  $H$  with degree at least  $4k$  is at most  $\frac{v(H)}{2}$ .*

**Proof.** (i) follows directly from the definition as we can order the vertices from left to right such that each vertex in  $H$  has at most  $k$  neighbors to its left. The number of edges is exactly the sum of the sizes of “left” neighbors of vertices of  $H$ . (ii) If there would be more than  $v(H)/2$  vertices with degree at least  $4k$ , the sum of degrees would be more than  $2kv(H)$  so the number of edges would be more than  $kv(H)$ , a contradiction to (i).  $\square$

The fact that algorithm  $FF$  uses at most  $O(k \log n)$  colors will be an immediate consequence of the following lemma:

**Lemma 2.2.** *For every integer  $j \geq 2$ ,  $|C_{4kj}| \leq |C_{4k(j-1)}|/2$ .*

**Proof.** for an integer  $i$ , let  $G_i$  denote the subgraph of  $G$  induced by the vertices of  $C_i$  (i.e., the subgraph induced by all vertices that are colored by  $FF$  with a color greater than or equal to  $i$ ). Note that  $C_{4kj} \subset C_{4k(j-1)}$ . Consider a vertex  $v \in C_{4kj}$ . Obviously,  $v$  is a vertex of  $G_{4k(j-1)}$ . It is easy to see that the degree of  $v$  in  $G_{4k(j-1)}$  is at least  $4k$ . This follows from the fact that at the time that  $v$  was revealed to algorithm  $FF$ , if  $FF$  assigned  $v$  a color  $c \geq 4kj$ , then  $v$  has a neighbor (in  $G_{4k(j-1)}$ ) that was already assigned a color  $i$  for every  $4k(j-1) \leq i < 4kj$ . By Lemma 2.1(ii), it follows that  $|C_{4kj}| \leq |C_{4k(j-1)}|/2$ . This completes the proof of the lemma.  $\square$

**Theorem 2.3.** *Let  $G = (V, E)$  be a  $k$ -degenerate graph and let  $c(V)$  be the coloring produced by the greedy algorithm  $FF$  on some input permutation  $\sigma$  of the vertices  $V$ . Then the number of colors used by  $FF$  is at most  $4k(\log n + 1) - 1$ .*

**Proof.** Using induction, Lemma 2.2 implies that

$$|C_{4k(\log n + 1)}| \leq \frac{|C_0|}{2^{1 + \log n}} = \frac{|C_0|}{2n}.$$

However  $|C_0| = n$ . So  $|C_{4k(\log n + 1)}| \leq \frac{1}{2}$ . Hence  $C_{4k(\log n + 1)} = \emptyset$ . Thus there are no vertices colored by  $FF$  with a color greater than  $4k(\log n + 1) - 1$ . This completes the proof of the theorem.  $\square$

## References

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